

The Case Study of a Computational Model for Immersive Human-Process Interactions within Design Optimization

Davide Guzzetti*
Auburn University

ABSTRACT

Virtual reality enables incorporating new cues and affordances into design optimization interfaces. Designers visually interact with optimization algorithms to examine solutions, define the search space, and present tradeoffs or outcomes. We hypothesize that additional dimensions of the human-process interface yield a higher probability of discovering the global optimal solution. To test this hypothesis, this work develops a computational model for visual-based process interactions in design optimization. The model is inspired by the Van Wijk model for visual analytics. Initial numerical results support the hypothesis.

Index Terms: Human-centered computing — Human computer interaction (HCI) — HCI theory, concepts and models; Human-centered computing — Visualization — Visualization theory, concepts and paradigms

1 INTRODUCTION

Design optimization is a recurring task in the creation of human artefacts, from the engineering of satellite systems to the construction of road bridges. Extensive research focuses on creating more accurate and faster gradient-based or meta-heuristic algorithms for design optimization. Nonetheless, human designers still heavily guide optimization algorithms via visual-based interactions to achieve optimal solutions. Minimal knowledge exists on how the nature and quality of such visual-based interactions impact how designers utilize algorithms (and algorithms inform designers) to generate optimal design solutions [1]. To characterize how the addition of new dimensions to a user interface impacts the probability of discovering a global optimal solution, this work presents an initial computational model for visual-based human-process interactions in design optimization. The model is initially applied to a canonical optimization test function, demonstrating a positive correlation between the optimality of the solution and the field of regard of the human-process interface.

2 MODEL

Inspired by the Van Wijk model for visual analytics [2], we developed an actionable computational model to render visual-based human-process interactions in design optimization. The interaction process is summarized in Figure 1, which serves as a reference for this section. Design space exploration parameters (block 6) at the current cycle, i , are rendered by a multivariate random variable, $\mathbf{p}_i$, which is drawn from a known multivariate probability distribution $\mathbf{f}_p$ parametrized in the vector $\mathbf{a}_i$.

*guzzetti@auburn.edu

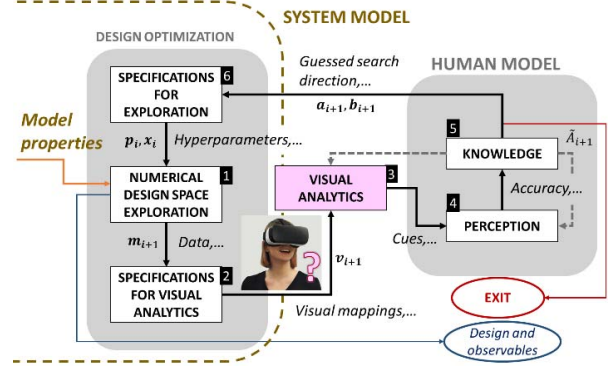


Figure 1: Human-process interaction model.

Let $\mathbf{x}_i$ be the current guess for the optimal solution. This guess is drawn, $\mathbf{x}_i \sim \mathbf{f}_x(\mathbf{b}_i)$, from a known multivariate probability distribution $\mathbf{f}_x$ parametrized in the vector $\mathbf{b}_i$. Let $\mathbf{x}^*$ denote a desired optimal solution. At block 1, estimate the optimal solution $\mathbf{x}_{i+1} = \arg\min_{\mathbf{x}_i, \mathbf{p}_i} y(\mathbf{p}_i)$ for the single objective function $y(\mathbf{p}_i)$, design space exploration parameters, $\mathbf{p}_i$, and initial guess $\mathbf{x}_i$. The numerical exploration of the design space produces a vector of observables, $\mathbf{m}_{i+1}$. The vector of observables can include figures of merit such as the error relative to the true optimal solution. At block 3, observables, $\mathbf{m}_{i+1}$, are presented to the user according to visualization specifications and affordances encoded by a vector, $\mathbf{v}$. The sets of observables in the last n cycles and visualization properties, $\mathbf{v}$, determine the perceived cognitive effort, $\tilde{E}_{i+1} = g_E(\{\mathbf{m}_j\}_{i+1-n}^{i+1}, \mathbf{v}_{i+1})$, solution accuracy, $\tilde{A}_{i+1} = g_A(\cdot)$, and time investment, $\tilde{T}_{i+1} = g_T(\cdot)$. In turn, perceived cognitive effort, solution accuracy, and time investment determine the exit conditions $e(\{\tilde{E}_j\}_{i+1-n}^{i+1}, \{\tilde{A}_j\}_{i+1-n}^{i+1}, \{\tilde{T}_j\}_{i+1-n}^{i+1}, \{\tilde{E}_{i+1}\})$, which, if true, terminate the procedure. Otherwise, if all exit conditions are false, the probability distribution for the design space exploration, $\mathbf{a}_{i+1} = \mathbf{u}_a(\tilde{E}_{i+1}, \tilde{A}_{i+1}, \tilde{T}_{i+1}, \mathbf{a}_i, \mathbf{p}_i)$, and probability distribution for optimal solution guess, $\mathbf{b}_{i+1} = \mathbf{u}_b(\tilde{E}_{i+1}, \tilde{A}_{i+1}, \tilde{T}_{i+1}, \mathbf{b}_i, \mathbf{p}_i)$, are updated. The cycle runs until the exit conditions are met. Elucidating the impact of user interface affordances on optimization outcomes is accomplished by characterizing $\mathbf{m}^*|\{\mathbf{v}\}$, where $\mathbf{m}^*$ are the process observables for the best solution discovered, and $\mathbf{v}$ is the independent variable. The procedure above can be implemented numerically and the relationship $\mathbf{m}^*|\{\mathbf{v}\}$ can be estimated via Monte Carlo simulation.

3 CASE STUDY

This section discusses an example application of the proposed model for human-process interactions in optimization. Start

considering the Rastrigin function $y(x) = An + \sum_{j=1}^n [x_j^2 - A \cos(2\pi x_j)]$ which has a global minimum $f(0, \dots, 0) = 0$. The Rastrigin function is a well-known test function for single-objective optimization, due to its multimodal nature. Despite its simplicity, this function displays a complex basin of convergence. For simplicity of exposure, let's reduce the test function to one dimension. Let's also assume that the human designer directs the search for a global optimum by setting the boundaries of the search, that is $x^* = \operatorname{argmin}_{x \in [b_l, b_u]} y$. Label the interval $[b_l, b_u]$ as the seeker. Assume that the human operator guesses the optimal solution as $x_i \sim \mathcal{N}(\mu_i, \sigma_i)$, where $\mathcal{N}$ is a normal distribution with mean μ_i and standard deviation σ_i . Also assume that the boundaries are defined as $(b_l)_i = \mu_i - \sigma_i$ and $(b_u)_i = \mu_i + \sigma_i$. While it is not necessary to constrain the optimum search to a finite interval for such a simple problem, this assumption mimics the common practice of defining exploration boundaries in real-world optimization design problems. The optimization problem is solved numerically using MATLAB *fmincon* with default tolerances, the prescribed search boundaries, and selecting sequential quadratic programming as algorithm. At each optimization cycle the boundaries of the search are updated toward the true optimal solution according to $\mu_{i+1} = \mu_i + \epsilon^\mu e_x$, where $e_x = x^* - x_{i+1}$ and $\sigma_{i+1} = \sigma_i + \epsilon^\sigma$. The first equation describes the tendency of the seeker central point to move toward the optimal solution at a rate ϵ^μ . The second equation describes the change in attitude toward exploiting the current solution ($\epsilon^\sigma < 0$) or exploring the solution space ($\epsilon^\sigma > 0$).

Next, the rate at which the seeker central point tends toward the global optimum, ϵ^μ , is written as a function of the number of **disjoint data representations (DDR)**s that the operator has to examine. A disjoint data representation is a representation that requires context switching. In particular, the relationship between the rate ϵ^μ and the number of DDRs to be examined by the designer, v , is modelled as follows

$$\epsilon^\mu = \begin{cases} 1 & \text{for } v < 3 \\ 1 - (v-3)/8 & \text{for } 3 \leq v \leq 7 \\ 0.5 e^{-c(v-7)} & \text{for } v > 7 \end{cases}$$

If the number of DDRs is less than 3, it is reasonable to assume immediate cognition and immediate identification of the optimal solution. If the number of DDRs is larger than 3 but lower than 7 (the average capacity of short-term memory [3]), the search rate is rendered by a linear relationship with the number of DDRs. If the number of DDRs surpasses the average STM limit, the search rate is assumed to be exponentially smaller with the number of DDRs. The number of DDRs is computed as follows $v = \frac{s}{q} C_r = \frac{1}{s} \frac{q!}{r!(q-r)!}$ where r is the dimension of the data representation, q is the dimension of the data, and s is the number of joint (i.e., simultaneous) visualizations. For two-dimensional plots ($r=2$) of 10-dimensional data ($q=10$), there exist ${}_{10}C_2 = 45$ possible plots. If the field of regard of a general computer monitor, accommodates $s=3$ visualizations, there are 15 DDRs ($v = 15$). On the contrary, if the field of regard of a VR application accommodates $s=9$ visualizations, there exist 5 DDRs ($v=5$). The exploration/exploitation tendency ϵ^σ is negative (i.e., exploitation) if the optimality of the solution improves, and positive (i.e., exploration) if the optimality of the solution does not improve. The exit conditions are defined as a stochastic function of the perceived time devoted to the task and the perceived accuracy of the solution. The probability to exit the optimization cycle due to perceived time runout is estimated as $P_{\text{end},1} =$

$\min\left(S_{0.5,10}\left(\frac{\sum n_i^{\text{eval}}}{n_{\text{max}}^{\text{eval}}}\right), 1\right)$, where the sigmoid function S is defined as $S_{0.5,10}(x) = \frac{1}{1+e^{-10(x-0.5)}}$ and n_i^{eval} is the number of function evaluations over a maximum tolerated. The probability to exit the optimization cycle due to perceived accuracy stagnation is described as $P_{\text{end},2} = \min\left(S_{0.5,10}\left(\frac{\sum n_i^{\text{imp}}}{n_{\text{max}}^{\text{imp}}}\right), 1\right)$ where n_i^{imp} is the number of optimization cycles since the last improvement in the objective function value. The total exit probability is calculated as the combined probability of a time runout or accuracy stagnation exit event. At each optimization cycle, the objective function value, y , is recorded. The final objective function value is tested against the variation of the DDRs number, v . In particular, a value $v=15$, representing a flat-screen interface, is compared to a value $v=5$, representing a VR interface. We execute 5000 Monte Carlo trials (recall, the model is stochastic) and collect 5000 trajectories for the objective function value. Figure 2 displays the average trend across the 5000 trajectories and corresponding standard deviation (std). Figure 2 reports one of three independent Monte Carlo analyses. The trend in Figure 2 indicates that (1) on average, the process that includes the VR interface (i.e., lower number of DDRs) reaches the optimal solution faster; and (2) on average, the process that includes the VR interface (i.e., lower number of DDRs) reaches a solution closer to the global optimum. Additionally, this example supports the hypothesis that optimality is impacted by affordances of the visual representation when human-process interactions are considered.

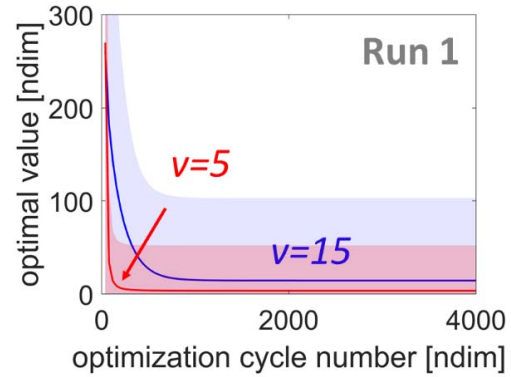


Figure 2: Monte Carlo simulation results.

4 CONCLUSION

Using a canonical optimization test function as case study, initial numerical experiments display a positive correlation between the dimensionality of the human-process interface and the quality of the final optimal solution.

REFERENCES

- [1] S. Ghosh and B. W. Bequette, "Process systems engineering and the human-in-the-loop: the smart control room," *Industrial & Engineering Chemistry Research*, vol. 59, no. 6, pp. 2422–2429, 2019.
- [2] J. J. Van Wijk, "The value of visualization," in *VIS 05. IEEE Visualization*, 2005, pp. 79–86.
- [3] R. Schweickert and B. Boruff, "Short-term memory capacity: Magic number or magic spell?," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 12, no. 3, p. 419, 1986.